

Name : Prathmesh Pakhale

Roll No. b4 41

```
COCOMO_MODES = {
    "1": {"name": "Organic", "a": 2.4, "b": 1.05, "c": 2.5, "d": 0.38},
    "2": {"name": "Semi-Detached", "a": 3.0, "b": 1.12, "c": 2.5, "d": 0.35},
    "3": {"name": "Embedded", "a": 3.6, "b": 1.20, "c": 2.5, "d": 0.32},
}

def calculate_cocomo():
    print("--- Basic COCOMO Estimator ---")
    print("Choose Project Mode:")
    print("1. Organic (Small teams, flexible requirements)")
    print("2. Semi-Detached (Medium teams, mixed experience)")
    print("3. Embedded (Tight constraints, highly complex)")

    mode_choice = input("Enter choice (1-3): ").strip()

    if mode_choice not in COCOMO_MODES:
        print("Invalid selection. Exiting program.")
        return

    try:
        kloc = float(input("Enter project size in KLOC (Kilo Lines of Code): "))
        if kloc <= 0:
            print("KLOC must be greater than 0.")
            return
    except ValueError:
        print("Invalid number entered for KLOC.")
        return

    mode = COCOMO_MODES[mode_choice]
    a, b, c, d = mode["a"], mode["b"], mode["c"], mode["d"]

    effort = a * (kloc**b)
    time = c * (effort**d)
    staff = effort / time

    print("\n--- Estimation Results ---")
    print(f"Selected Mode: {mode['name']}")
    print(f"Project Size: {kloc:.2f} KLOC")
    print(f"Effort Required: {effort:.2f} Person-Months")
    print(f"Development Time: {time:.2f} Months")
    print(f"Persons Required: {staff:.2f} People")
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if __name__ == "__main__":  
    calculate_cocomo()
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Output

--- Basic COCOMO Estimator ---
Choose Project Mode:
1. Organic (Small teams, flexible requirements)
2. Semi-Detached (Medium teams, mixed experience)
3. Embedded (Tight constraints, highly complex)
Enter choice (1-3): 1
Enter project size in KLOC (Kilo Lines of Code): 3

--- Estimation Results ---
Selected Mode:    Organic
Project Size:     3.00 KLOC
Effort Required:  7.61 Person-Months
Development Time: 5.40 Months
Persons Required: 1.41 People
```

```
Output

--- Basic COCOMO Estimator ---
Choose Project Mode:
1. Organic (Small teams, flexible requirements)
2. Semi-Detached (Medium teams, mixed experience)
3. Embedded (Tight constraints, highly complex)
Enter choice (1-3): 2
Enter project size in KLOC (Kilo Lines of Code): 30

--- Estimation Results ---
Selected Mode:    Semi-Detached
Project Size:     30.00 KLOC
Effort Required:  135.36 Person-Months
Development Time: 13.93 Months
Persons Required: 9.72 People
```

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```
--- Basic COCOMO Estimator ---
Choose Project Mode:
1. Organic (Small teams, flexible requirements)
2. Semi-Detached (Medium teams, mixed experience)
3. Embedded (Tight constraints, highly complex)
Enter choice (1-3): 3
Enter project size in KLOC (Kilo Lines of Code): 30

--- Estimation Results ---
Selected Mode:    Embedded
Project Size:     30.00 KLOC
Effort Required:  213.23 Person-Months
Development Time: 13.91 Months
Persons Required: 15.33 People
```